Supplementary Material
Verify, Look Again, Remember: Fail-Closed Vision-Language Checks and Safety-Monotone Memory for Relief Delivery Robots

Anonymous Authors

Code, data and every trial log: https://github.com/fe2131-art/rescue-mind

This document holds material moved out of the main paper. Section, table and figure numbers with an S prefix refer to this document; unprefixed references are to the main paper. All numbers are generated from the same result files as the main paper. It is organised as follows. *Methods and background:* S1 development data, T1 construction and the two-model rule; S2 model selection on development data; S3 the extended related work. *Results:* S4 additional T1 analyses (decision-rule ablations, risk–coverage, weaker models); S5 campaign E details; S6 wall-clock timing; S7 further result details (studies Z, P, R, campaigns F, G and H, hazard fixes). *Failure analysis:* S8 lessons from running the real stack. *Audit:* S9 the registration audit and threats to validity.

S1. Development Data, T1 Construction and the Two-Model Rule

Development data. Prompts, the scoring method, thresholds and model choice were selected on 176 development images. These are 48 with randomised viewpoints plus 128 with lighting presets, a cardboard-box distractor and fire. The instruction parser was tuned on 40 development utterances. The later hazard fixes (main paper, Sec. V-C) also used two hazard development sets of 300 rendered images each: DH, and DH2, which emphasises small early-stage fires.

Two-model rule. The two-model rule (main paper, Sec. III-C) approves only if Eq. (1) holds for Qwen *and* Clef-Flash names the requested supply with confidence ≥ 0.65, presence ≥ 0.80 and hazard probability below its own threshold.

T1 examples. Fig. S1 shows held-out T1 images. T1 used 60 Gazebo worlds $\times$ 2 stations $\times$ 5 viewpoints with continuous random lighting, supplies moved by up to ± 0.15 m and freely rotated, random fire positions, and three distractors. Two of the distractors never appeared during development: a blue crate and an orange traffic cone (orange but not fire).

S2. Model Selection on Development Data

Models were ranked lexicographically. The first key was zero false approvals and zero unsafe intent acceptances, and the second was explanation and intent accuracy. Qwen3-VL-8B parsed more development utterances but accepted two unsafe ones. Qwen3.5-4B produced a false approval. Qwen3.5-9B was therefore chosen and frozen before the T1 and E registrations.

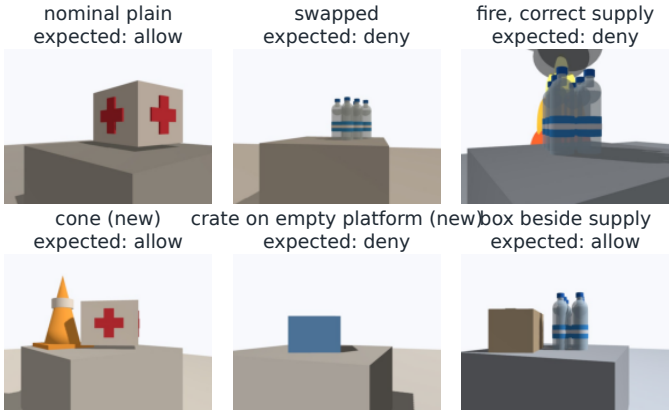


Fig. S1. Held-out T1 images (robot camera, 640×480) with the generator's expected decision. The crate and the traffic cone never appeared during development.

TABLE S1
Candidate models with the final prompts (`v35`: checkpoint questions of v3.5; `i35`: instruction parser) on development data only. Images: correct decisions with false approvals (FA) and false denials (FD). Explanations: one-sentence reports judged correct against the generator labels. Intent: development utterances, with unsafe acceptances. VRAM in MiB.

Model	Released	Images (176)	Explanations (128)	Intent dev (40)	VRAM
Qwen2.5-VL-3B	2025-01	176/176 (FA 0/FD 0)	84/128	31/40, unsafe 0	–
Qwen3-VL-8B	2025-10	176/176 (FA 0/FD 0)	120/128	38/40, unsafe 2	17212
Qwen3.5-4B	2026-02	175/176 (FA 1/FD 0)	125/128	35/40, unsafe 2	9102
Qwen3.5-9B	2026-02	176/176 (FA 0/FD 0)	125/128	35/40, unsafe 0	18320

S3. Extended Related Work

The main paper condenses this section; the full text follows.

Failure detection and verification with foundation models. REFLECT [1] and AHA [2] explain manipulation failures in natural language after the fact. Inner Monologue [3] feeds success detection back into a planner. Du et al. [4] fine-tune a VLM as a success detector. Sentinel [5], FAIL-Detect [6] and FIPER [7] monitor learned policies at run time

(FIPER calibrates its out-of-distribution and uncertainty scores conformally on successful rollouts and requires both signals to alarm); Code-as-Monitor [8] checks task constraints with VLM-generated programs, both reactively and proactively, without a calibrated threshold or a ground-truth audit; an LLM-based anomaly monitor triggers reactive fallbacks [9]; and uncertainty-aware policy steering decides when to act, ask or learn [10]. *Closest prior art.* VAP-TAMP [11] (arXiv preprint, April 2026; we read the full text) verifies the preconditions and effects of each PDDL action with a VLM. For each predicate it generates several semantically equivalent yes/no questions, takes a majority vote and, if the answers agree, asks the VLM whether the view is sufficient; otherwise the VLM proposes a viewing direction and the robot moves there, up to a viewpoint budget. A failed verification corrects the scene graph and triggers PDDL replanning. It was evaluated by task success on 100 real mobile-manipulation trials and on injected failures in OmniGibson. The overlap with our design is real: paraphrase agreement is the counterpart of our permuted-question agreement, and VLM-chosen views are the counterpart of our look-around. To our knowledge, the differences are (i) our scores are next-token probabilities over a closed answer set with a threshold calibrated on development data, whereas VAP-TAMP uses textual yes/no answers and a vote; (ii) in our design every check is combined in a fail-closed conjunction, so a failed check or an extra view can only refuse or revoke, whereas in VAP-TAMP a verification result corrects the symbolic state in either direction; (iii) an independent velocity guard that the VLM cannot command; (iv) memory that may only reorder checks and add caution; and (v) preregistered closed-loop audits of the error direction against simulator ground truth, which report the unsafe-delivery rate and not task success. VAP-TAMP covers things we do not: PDDL task and motion planning, manipulation, and real-robot trials. Semantic anomaly detection with LLMs [12] and visual question answering that abstains rather than answers wrongly [13] are close in spirit to a gate that prefers refusal. Our check runs *before* the robot acts and is closed-set and zero-shot. Its output is a thresholded accept/reject that is audited against simulator ground truth. The closest methodological relative is KnowNo [14], which scores multiple-choice answers by next-token probability and asks a human for help, with a conformal-prediction coverage guarantee (see [15] for background on conformal prediction). Introspective planning [16] extends this line by letting the planner reason about task ambiguity. Conformal risk control [17] controls the expected value of any monotone loss (for example a false-negative rate), and Learn then Test [18] gives high-probability risk control; our study K (held-out fire miss rate $\leq \alpha$ with probability 0.9) is of that form. Conformal decision theory [19] calibrates a robot decision rather than a prediction set. Our gate is a one-shot accept/refuse threshold and not an online-adaptive controller. SayCan [20] likewise scores a fixed candidate set by language-model likelihood. We share the scoring, but we respond to uncertainty by refusing or searching, and we add permuted-option agreement as a test-time defence against letter bias.

Multiple-choice scoring and its biases. Scoring answer symbols works well when a model binds symbols to options [21]. Language models are reasonably calibrated on well-formatted multiple-choice questions [22], but they prefer certain option IDs [23] and orders [24], and their raw confidence needs validation [25], [26]. On embodied question answering, even the best frontier model reaches only 42.79% recall when it should abstain [27], which supports a calibrated refuse option over trusting the model to abstain. Our two permuted questions are a minimal, deterministic form of self-consistency [28]. The Yes/No hazard question follows the polling style of POPE [29].

Safety architectures. Simplex [30], [31], shields [32] and control barrier functions [33] keep a simple verified layer between a complex decision maker and the actuators. LLM-controlled robots can be manipulated into harmful actions [34], and guardrails such as RoboGuard [35] check plans before execution. The safety filter [36] is the current umbrella for such least-restrictive overrides of a nominal controller; our velocity guard is a minimal instance and the VLM check is a perception-side gate in the same spirit. Closest in using a VLM for safety, Santos et al. [37] turn images and language feedback into a failure set that a Hamilton–Jacobi safety filter enforces, so the guarantee comes from the verified filter; ours is a union bound over check errors. RoboSafe [38] guards VLM-driven embodied agents with executable safety rules and a short- and long-term safety memory, which drives rule-based replanning and is not constrained to be safety-monotone. In contrast to vision-language-action models [39], [40], our VLM never commands motion.

Memory for robots in changing environments. Robots increasingly keep long-horizon semantic memory. ReMEmbR [41] and Embodied-RAG [42] retrieve from a database of captioned observations; KARMA [43] adds long- and short-term memory to embodied agents, and H-EMV [44] verbalises a robot's episodic memory. 3D scene graphs such as Hydra [45], ConceptGraphs [46] and HOV-SG [47] support planning [48] and language grounding [49]. Long-term mapping has long dealt with change [50], [51], and recent work tracks changes in dynamic scene graphs [52], [53]. These systems use memory to *answer* or *plan*. A recent preprint reports, in a small grid-navigation testbed, that acting on unaudited stale memory can be worse than having no memory [54]. Long-term agent memory is also an attack surface: corrupted entries can steer an LLM agent [55] (an LLM-agent result, not a robotics one). Memory benchmarks such as FindingDory [56] measure how well memory supports recall in long-horizon tasks, not whether acting on it is safe once the world has changed. GLOVE [57] and VLM-DEWM [58] check memory or a world model against fresh observations to keep the memory *correct*.

Object search with priors and semi-static maps. Using past observations to decide where to look first is a classic idea, and we do not claim it as ours. Active visual object search keeps a belief over target locations and plans where to look next, using uncertain spatial and semantic

knowledge [59], inter-object relations in Semantic Linking Maps [60], or object-oriented POMDPs whose belief factorises over objects [61]. Long-term mapping models objects or features that persist, appear or disappear (semi-static) and detects change [62], [63]; DovSG updates an open-vocabulary 3D scene graph locally as objects are moved [64]. Zero-shot object navigation with language or vision-language priors, such as ESC [65] and VLFM [66], uses a pre-trained model as a prior over where to explore, but has no mission-to-mission memory of a specific site. To our knowledge, these works optimise search efficiency (path length, success rate, map accuracy) and treat a detection, or an updated map, as the basis for the next action. We did not find in them a bound on the unsafe-action risk caused by *stale* memory at a pre-action gate. Our memory is narrower and serves a different goal: it may only reorder stations and add caution, the on-site verification is never skipped or relaxed, and this restriction is enforced in code and checked by the run audit, which gives a union bound that does not depend on how old the memory is. These methods could supply the ordering prior in our setting, for example a belief over supply location, provided the gate stays unchanged; combining them is future work, and we have not compared search efficiency against them. We study a different use: memory decides *where and how carefully to verify*. We restrict it so that it can never reduce verification, and we state this as a provable invariant and test it under scripted change.

Active perception. Moving the sensor to resolve ambiguity is a classic idea [67], [68], and view planning for inspection is well studied [69]. Recent embodied agents explore until a VLM is confident [70] or let a VLM choose views [71]. Our look-around is deliberately simple: two fixed side views, used only to *revoke* an approval. This makes its effect auditable and its safety direction obvious.

Robustness and the sim-to-real gap. Image classifiers degrade under common corruptions [72], and so do detectors [73], adapted vision-language models [74] and video-language models [75]. Sim-to-real transfer is a recognised challenge in robotics [76]. Without a physical platform, we measure one part of this gap, sensor degradation, under a pre-registered protocol. What we care about is the *direction* of the errors, not only accuracy.

Diverse redundancy. Requiring agreement between independently developed versions is a classic fault-tolerance technique [77]. Its benefit is limited by correlated failures [78], and model-agreement schemes for language models have reappeared recently [79]. We report both sides: the two models agree on most images, and the second model's value depends strongly on how it is asked.

Evaluation practice. We follow recommendations to report conditions, multiple metrics, uncertainty and failure modes [80]. We use Wilson intervals [81] and exact McNemar tests [82]. Domain randomisation [83] is used here to build a held-out *test* distribution, not for training. Our design is fixed-n and pre-registered; sequential testing with valid early stopping [84] is a possible refinement for small n. How far simulation predicts real-world behaviour is itself measurable [85]; we make no claim beyond simulation.

Fire and smoke perception. We use D-Fire [86] for study P because it is CC0 and ground-level with negative images; FASDD [87] is a larger alternative with ground-level, UAV and satellite subsets and a possible future source. Local open-weight VLMs have been used for wildfire detection from UAV imagery [88], descriptively and without refusal semantics or a closed-loop audit.

TABLE S2
DECISION-RULE ABLATIONS ON T1: FALSE APPROVALS / FALSE DENIALS. *: EXACT MCNEMAR $p < 0.05$ VERSUS THE FULL RULE OF THE SAME MODEL.

Decision rule	Qwen3.5-9B	Qwen3.5-4B	Qwen3-VL-8B	Qwen2.5-VL-3B
Full rule	5 / 4	9 / 1	4 / 8	4 / 4
A1: question 1 only	5 / 4	9 / 0	4 / 7	4 / 4
A2: question 2 only	5 / 1	9 / 1	4 / 2*	36 / 1*
A3: no hazard question	47 / 4*	48 / 1*	46 / 8*	48 / 4*
A4: no threshold (τ_s=0)	5 / 4	9 / 1	4 / 8	4 / 4
v3.5: τ_h=0.05 (I-10)	3 / 4	4 / 1	4 / 8	1 / 30*

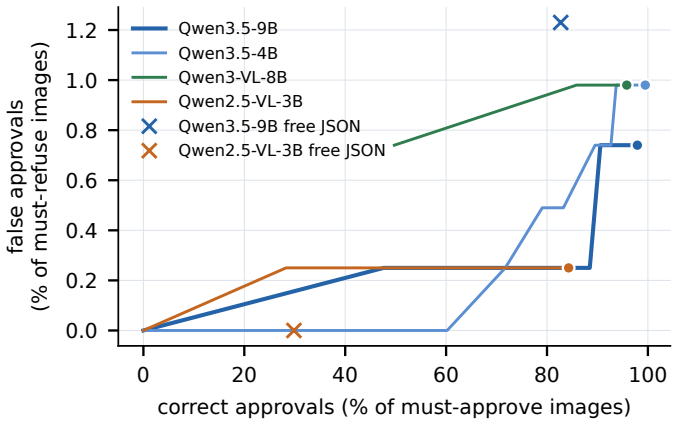


Fig. S2. T1 risk–coverage: false-approval rate versus correct-approval rate as τ_s varies ($\tau_h = 0.05$, v3.5). Markers show the frozen $\tau_s = 0.5$. The curves are flat over most of the range: the remaining errors are missed fires, which τ_s cannot fix.

S4. ADDITIONAL T1 ANALYSES

A. Decision-rule ablations

Table S2 isolates the parts of the rule on T1 (Table II, Eq. 1). For the 9B model neither the permuted second question nor the confidence threshold changed the number of false approvals. With Qwen2.5-VL-3B, requiring the two multiple-choice questions to agree reduced false approvals, which is the option-letter bias reported in [23].

B. Risk–coverage

Fig. S2 varies τ_s only for description, because τ_s was not tuned on T1.

C. Weaker models and free-form output

For Qwen2.5-VL-3B the second question alone approved 36 must-refuse images, and requiring agreement with the first question brought this down to 4, which is the option-letter bias reported in [23] in action (Table S2). Asking this model for free JSON (A5) left 148 of the 1200 generations unparseable

TABLE S3
CAMPAIGN E (RANDOMISED WORLDS) BY STRATUM.

Expected	Condition	n	Correct	Observed
allow	box	2	2	allow 2
allow	cone	4	4	allow 4
allow	crate	2	2	allow 2
allow	plain	12	12	allow 12
deliver after search	box	2	2	allow after search 2
deliver after search	crate	2	2	allow after search 2
deliver after search	plain	12	12	allow after search 12
deny	box	2	2	deny 2
deny	crate	4	4	deny 4
deny	plain	6	6	deny 6
deny hazard	fire	8	8	deny hazard 8
deny hazard after search	fire other	4	4	deny hazard after search 4

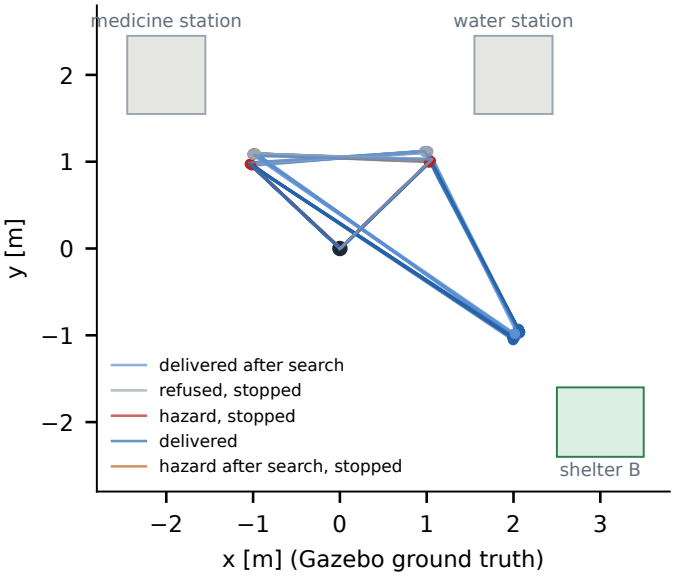


Fig. S3. Campaign E: ground-truth trajectories of all final trials, coloured by audited outcome.

and approved 57 images; Qwen3.5-9B gave no parse errors (main paper, Sec. V).

S5. CAMPAIGN E DETAILS

Campaign E used a new randomised world in every trial (main paper, Sec. V); Table S3 gives the outcomes by stratum and Fig. S3 the ground-truth trajectories.

S6. WALL-CLOCK TIMING

Where the time goes. Table S4 breaks down the 34 delivered missions that needed no search. The check itself takes about 5.5 s: aiming, waiting until stationary, frame checks, and the scored VLM queries (3.4 s). Driving between station and shelter takes about 39 s (sum of phase medians). The check is therefore short compared with driving. The situation report adds about 5 s but could run in parallel with the drive to the shelter.

S7. FURTHER RESULT DETAILS

Study Z. The contrastive baselines used SigLIP [89] and CLIP [90] prompts for the supply and the hazard, and the same crops as our check. On development data no threshold pair gave both few false approvals and many correct ones. The contrastive scores do not separate "right supply, no fire" from the alternatives in these scenes.

TABLE S4
WALL-CLOCK PHASES OF DELIVERED MISSIONS WITHOUT SEARCH (CAMPAIGNS D AND E; V3.4 AND V3.5, NO LOOK-AROUND).

Phase	Median [s]	Min	Max
drive to station	14.31	12.19	16.62
aim camera	0.50	0.50	0.52
wait until stationary	0.67	0.66	0.71
acquire and check frames	0.90	0.75	0.98
VLM decision (scored queries)	3.44	2.95	6.64
situation report (not used to decide)	5.14	3.99	8.54
drive to shelter	24.83	15.02	25.32

T1 hazard-question false alarms. With the registered v3.4 rule, the hazard question raised false alarms on 0/25 traffic-cone images and on 0/523 images without fire. With the v3.5 rule, it raised false alarms on 0/25 cone and 0/523 fire-free images.

Campaign F, remaining misses. Six of v3.6's seven remaining misses were fires of 1 100–2 400 visible pixels with hazard probabilities of 0.002–0.006, which overlap those of fire-free views, so no threshold can separate them. The seventh showed 8 600 pixels at $p_h = 0.019$.

Campaign G, world set and medians. The set of worlds is defined from mask renders at all three views, so it includes the side views by construction. The medians over all delivered runs, 102 s and 122 s, are not comparable, because look-around refused most of the fire runs that the front-only system delivered.

Study P, false alarms and licence. D-Fire [86] was released by its authors under CC0, as stated in the original repository's LICENSE; we used a Hugging Face mirror. Rendered fire is stylised, which motivated this check on real photographs. D-Fire's fire-free class comes from fire-surveillance cameras and is rich in hard negatives: hazy hillsides, low clouds, a smudge on the lens, red sunset clouds, glowing tents. Everyday scenes such as streets, beaches and roads scored $p_h \approx 0.01$. The question was worded to catch faint smoke ("a grey or black cloud") and thresholded at 0.05, so it errs strongly toward stopping on real images. We registered the site-specific threshold calibration as a follow-up study (P2).

Campaign H, failure types. Staleness predicts two failure types for naive memory: delivering a supply that has been moved away, and delivering where a fire has appeared since the last visit; the per-trial ledger records which occurred.

Study R, all conditions. Table S5 gives the false approvals, correct approvals and acquisition-gate rejections of study R (main paper, Sec. V-D) for every degradation and severity.

Hazard fixes on development data. Table S6 compares the candidate hazard fixes of the main paper (Sec. V-C).

Campaign H, per-mission outcomes. Fig. S4 shows every mission of campaign H (main paper, Sec. V-E).

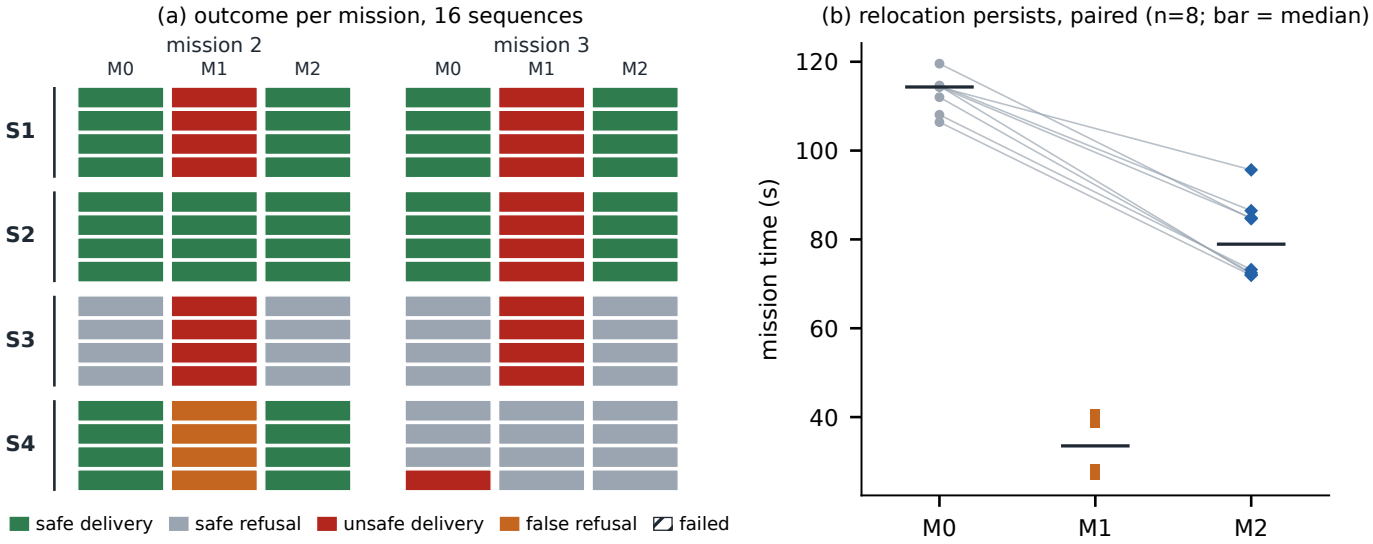


Fig. S4. Campaign H, missions 2 and 3 of the 16 sequences. (a) Outcome of each mission under no memory (M0), naive memory (M1) and verified memory (M2); rows are sequences, grouped by scripted story (S1–S4; definitions in the caption of Table S8). No mission failed (the hatched "failed" category is empty). (b) Mission time of the eight missions whose relocation persisted, for each arm; grey lines join the M0 and M2 times of the same mission, bars mark medians. M1 times are short because M1 skips the checks.

TABLE S5
STUDY R: V3.6 OFFLINE CHECK UNDER CAMERA DEGRADATIONS (T2 SUBSET). SEVERITY M = MODERATE, S = SEVERE.

Degradation	Sev.	False appr.	Fire missed	Correct appr.	Gate rej.
None (clean)	–	0/192	0/27	107/108	0
Sensor noise	M	0/192	0/27	107/108	0
Sensor noise	S	0/192	0/27	107/108	0
Motion blur	M	0/192	0/27	106/108	0
Motion blur	S	0/192	0/27	101/108	0
Defocus	M	0/192	0/27	108/108	0
Defocus	S	0/192	0/27	108/108	0
Under-exposure	M	0/192	0/27	107/108	0
Under-exposure	S	0/192	0/27	106/108	0
Over-exposure	M	0/192	0/27	96/108	4
Over-exposure	S	0/192	0/27	59/108	24
JPEG	M	0/192	0/27	108/108	0
JPEG	S	0/192	0/27	107/108	0

TABLE S6
HAZARD FIXES ON DEVELOPMENT DATA (DEV V1+V2, DH, DH2). $\tau_h = 0.05$ UNLESS STATED.

Configuration	False appr.	Correct appr.	Fires refused	One V100
v3.5: Qwen3.5-9B	46/413	357/360	165/211	yes
+ enlarged crops	37/413	356/360	174/211	yes
+ hidden-fire wording	25/413	357/360	186/211	yes
+ wording + crops (v3.6)	17/413	357/360	194/211	yes
Qwen3.5-4B	37/413	360/360	174/211	yes
Clef-Flash, original Jev request	45/413	182/360	166/211	yes
Clef-Flash, task wording	42/413	350/360	169/211	yes
Clef (task) + Qwen (v3.6), both approve	15/413	350/360	196/211	no

Campaign H, case study. Fig. S5 shows the one mission in which M0 delivered into a fire (sequence Q09, mission 3).

Memory caution versus a global threshold (descriptive, not registered). Campaign H stored decisions, not every view's p_h, so only a partial comparison is possible from existing logs. *Fires:* the one mission where caution changed an outcome is Q09.3 (M0 $p_h = 0.035$, M2 0.049); a global $\tau_h = 0.0113$ (study K, $\alpha = 0.10$) would have stopped M0 there too, since $0.035 > 0.0113$. *False stops:* on the 193 fire-free T2 views the global $\tau_h = 0.0113$ gave 4/193 false alarms (2.1% per view, about 6% per approval with three views if independent), against 0/193 at the deployed 0.05. Caution applied in 12 of the 32 M2 missions: 8 at stations with a fire (all 8 stopped) and 4 at fire-free stations, with 0 false refusals there (Wilson upper bound 49%). That is compatible with both zero and the global rate, so a comparison at equal false-stop rate cannot be made from these logs. What caution does buy by construction is lower exposure: the strict threshold applies at stations with a recent hazard, not everywhere. A direct test needs many fire-free missions at stations with a hazard history; we did not run one.

TABLE S9
SECOND-MODEL OVERLAP ON THE 211 VISIBLE FIRES OF THE DEVELOPMENT SETS DH AND DH2 (NO TEST DATA). A FIRE IS MISSED WHEN THE CHECKER WOULD APPROVE THE VIEW (QWEN3.5-9B V3.6, $\tau_h = 0.05$; CLEF-FLASH WITH TASK WORDING, HAZARD THRESHOLD 0.05; SAME RULES AS TABLE S6). SCRIPT: SCRIPTS/JEV_OVERLAP.PY.

	Clef-Flash misses	Clef-Flash catches
Qwen misses	15	2
Qwen catches	27	167

Qwen misses 17, Clef misses 42; P(Clef misses | Qwen misses) = 88% (Wilson [66, 97]%); Jaccard 0.34. Counting only the hazard question (probability ≥ 0.05 in the hazard question alone): Qwen misses 23, Clef misses 55, both miss 23.

A. *Campaign H: all paired missions*

Table S8 lists every paired mission of campaign H (missions 2 and 3 of the 16 sequences) with its outcome and mission time under M0, M1 and M2.

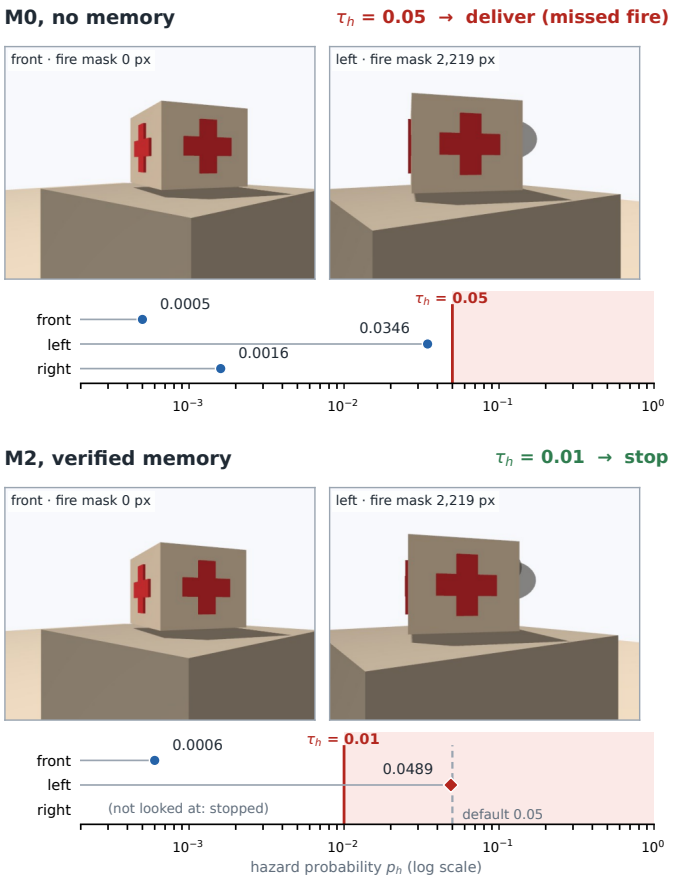


Fig. S5. Cross-mission memory (I-15), campaign H, sequence Q09 mission 3: the same world without memory (M0) and with verified memory (M2). Each row shows the front view and the left side view (fire-mask pixels from the simulator) and the hazard probability p_h of every view that was checked against the threshold (shaded: $p_h \geq \tau_h$). M0 used $\tau_h = 0.05$; the left view scored 0.035 and the robot delivered. M2 had seen a fire at this station two missions earlier, so it used $\tau_h = 0.01$; the left view scored 0.049 and the robot stopped without checking the right side. The fire is small and partly hidden from the left, so the two p_h values are close.

Repeated-mission model. The model of repeated missions, with the per-check rates measured in G and H ($p_{\text{det}} = 0.95$, $n = 44$), is shown in Fig. 5 of the main paper (Sec. V-E); its curves are computed from measured rates, so their order is a model prediction, not an observation. The figure also shows a fourth policy, M1age, a naive memory that is trusted only while its entry is recent (recency-gated, without verification); it lies between M1 and M2 in unsafe deliveries. Over the whole range of change probabilities $q \in [0, 0.5]$ the model finds no break-even point: M1 and M1age stay faster than M0, and M2 is faster than M0 at small q and converges to it as q grows, because M2 re-checks and so approaches the no-memory behaviour. At $q = 0.2$ per mission the unsafe deliveries per 100 missions are 35.83 for M1, 1.95 for M2 and 1.95 for M0, with mean mission times 32.4, 71.7 and 80.3 s. Neither the single missed-fire delivery nor the 32 paired missions measure how often a real world changes; q is a free parameter of the model.

More stations (descriptive, not registered). Our arena has $k = 2$ stations, so memory can save at most one search detour. Let the supply sit at one of k stations and move, with probability q between missions, to a uniformly chosen other station, so its long-run location is uniform. Without memory the robot visits stations in inventory order and needs $(k+1)/2$ visits on average; with verified memory it first visits the station where it last saw the supply and needs $1 + qk/2$. With the detour $\delta = 30$ s measured in H, the expected saving is $\delta\,((k-1) - qk)/2$ per mission and vanishes at $q^* = (k-1)/k$. For $k = 2$ this reproduces the Monte-Carlo model above (9 s at $q = 0.2$); for $k = 8$ it is 80 s. A simulation of the same process agrees within 1 s for $q > 0$ (`scripts/h_k_scaling.py`). The safety side does not depend on k in this way: each policy checks every station it delivers from with the same rule, so Corollary 1 applies unchanged, and memory lowers the expected number of checks per mission rather than their strength. The model assumes the stations are equally far apart and that supply identification is perfect (T1: no supply false approvals in 332 images).

S8. Failure Analysis and Lessons

Running the full stack on a GPU revealed 23 problems that 620 offline unit tests and mocked ROS runs had not; the most instructive are listed here (the full table is in `docs/FINDINGS.md` in the repository).

- **The auditor trusted the controller's sensor.** The wheel joint axis was expressed in the child-link frame after a 90° roll, so the wheels spun about the vertical axis. Odometry reported 1.5 m forward motion while the body slid sideways by 0.75 m. An audit based on `/odom` would have passed the run. Audits now use Gazebo's ground-truth odometry. Changing the wheel collision from a cylinder to a sphere reduced dead-reckoning drift on the medicine route from 0.16 to 0.011 m (worst of two repeats; `scripts/i14_pilot_p_range.py`).
- **Free-form JSON from the first VLM was unusable** on our renders. With Qwen2.5-VL-3B, 21 of 96 responses (48 images, two questions each; all 21 for the first question) could not be parsed and the model hallucinated water on empty shelves. A newer model parses reliably, but it still approves fewer correct deliveries than scoring does (Table II). This motivated closed-set scoring. T1 measured the free-JSON method again (Table II).
- **Middleware effects.** Best-effort QoS silently dropped 92 of 102 0.9 MB camera frames under load. The GPU lidar detected the robot's own camera mast. Slow DDS discovery on a 4-core host failed 4 of 12 starts in an aborted campaign. An interrupted supervisor left an orphaned stack that made every trial of the next campaign stop with two `/cmd_vel` publishers. All of these are recorded with their fixes; the aborted campaigns are published.
- **Stale-request latch.** Two pre-decision retries in campaign D happened while the robot was stationary: the mission node's request stream stalled for ≈0.3 s on the saturated CPU. We instrumented the mission loop, the guard and garbage collection and rejected two hypotheses (guard

starvation, GC pauses). Raising control-loop priority did not remove the stalls. We therefore changed the guard: a stale zero request now holds zero, while a stale motion request still latches. With a 0.6 s stall injected before departure, the old guard failed 2/2 runs and the new one completed 2/2. Raising priority with a file capability then had a side effect: it put the exec into the dynamic loader's secure mode, which stripped `LD_LIBRARY_PATH` and broke the ROS bag recorder. A regression run caught this before any campaign.

- **A pilot exposed a threshold that was not conservative enough.** Before campaign E we ran nine trials in separate pilot worlds. In one, the robot found the wrong item, searched the other station, and delivered from it although a fire burned behind it. The fire was almost entirely hidden by the first-aid kit, and $p_h = 0.32$. On development data no fire-free image exceeded $p_h = 0.0007$, so we lowered τ_h to 0.05 (v3.5). The same pilot worlds then gave 9/9 expected outcomes. T1 was not involved: both thresholds were registered for T1 before any T1 image was scored.
- **A generator bug caught by looking at the data.** In the first T1 render, distractors meant to stand next to a supply were placed through it. The emptiness check ran after the supplies were moved. The render was discarded and the fix recorded as a pre-registration amendment, before any model scored a T1 image. A second amendment replaced the field-of-view fire label with the rendered-mask label, after we saw fires hidden behind supplies that the geometric label had called visible.

S9. Registration audit

For every study with results, the registration commit precedes the results commit (holds). Campaign C's registration and ledger entered the repository together, so their order cannot be verified from the history.

A. Threats to validity

Internal. The studies were designed sequentially, each registered after the results of the previous ones were known, so registration is confirmatory within a study but not across the programme (Table S7); study K was registered after the T2 scores it reuses had been computed (`PREREGISTRATION_K.md`, offline, no new inference). The H stories are scripted so that a naive memory fails by construction; the M1–M2 contrast is therefore large by design, and the M2–M0 comparison is the one that bounds risk (`PREREGISTRATION_H.md`). The analysis scripts computed the median of an even-length list as the upper-middle element; this was fixed and all summaries regenerated, and errata with no change to protocols, endpoints or claims are in `PREREGISTRATION_{F,G,H}.md` (for H, a relocation-persist median of 79 s had been reported as 85 s). Campaign J deviated during the run (`PREREGISTRATION_J.md`): from trial J007 documentation builds shared the 4-core host, and alternating about 46 GB of weights through a 54 GB page cache made every check re-read its weights, so measured latency includes model I/O and is inflated in both arms; two first attempts failed before any VLM decision and were re-run under the registered retry rule.

Construct. Fire labels come from rendered mask pixels (a fire is perceivable at 614 pixels or more), not from human judgement, and the fire is stylised. "Delivery" means that the robot reached the shelter after approving; there is no handover, so grasping and placing are not tested.

External. All results use one arena geometry, two stations and simulation only. The real-photograph study (P) is descriptive and reports false alarms on D-Fire images; it does not test a robot. The two checkers in J share lineage (Clef-Flash is post-trained from Qwen3.5-9B), so errors may be correlated.

Statistical. Samples are small: P1 has 8 relocation-persist pairs, and P3 compares 0 unsafe deliveries (M2) with 1 (M0), which shows only that M2 is not less safe (Wilson upper bounds 11% and 16%). Images of one world, and missions of one sequence, are not independent; we therefore add world-cluster bootstrap intervals and a sequence-level McNemar test, which are wider or less powerful than the pooled ones. Several studies are reported without family-wise correction, because each has its own registered endpoint and the claims are not pooled.

References

[1] Z. Liu, A. Bahety, and S. Song, "REFLECT: Summarizing robot experiences for failure explanation and correction," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 3468–3484. [Online]. Available: https://proceedings.mlr.press/v229/liu23g.html

[2] J. Duan, W. Pumacay, N. Kumar, Y. R. Wang *et al.*, "AHA: A vision-language-model for detecting and reasoning over failures in robotic manipulation," in *International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/hash/70a06501001e1820fd1eb9ee821302d2-Abstract-Conference.html

[3] W. Huang, F. Xia, T. Xiao, H. Chan *et al.*, "Inner monologue: Embodied reasoning through planning with language models," in *Proceedings of The 6th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2022, pp. 1769–1782, coRL 2022 (PMLR volume dated 2023); arXiv preprint July 2022. [Online]. Available: https://proceedings.mlr.press/v205/huang23c.html

[4] Y. Du, K. Konyushkova, M. Denil, A. Raju *et al.*, "Vision-language models as success detectors," in *Proceedings of The 2nd Conference on Lifelong Learning Agents (CoLLAs)*, ser. Proceedings of Machine Learning Research, vol. 232. PMLR, 2023, pp. 120–136. [Online]. Available: https://proceedings.mlr.press/v232/du23b.html

[5] C. Agia, R. Sinha, J. Yang, Z. Cao *et al.*, "Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 689–723, coRL 2024 (PMLR volume dated 2025); method name: Sentinel. [Online]. Available: https://proceedings.mlr.press/v270/agia25a.html

[6] C. Xu, T. K. Nguyen, E. Dixon, C. Rodriguez *et al.*, "Can we detect failures without failure data? uncertainty-aware runtime failure detection for imitation learning policies," in *Robotics: Science and Systems XXI (RSS)*, 2025, method name: FAIL-Detect. [Online]. Available: https://doi.org/10.15607/RSS.2025.XXI.073

[7] R. Römer, A. Kobras, L. Worbis, and A. P. Schoellig, "Failure prediction at runtime for generative robot policies," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025, method name: FIPER.

[8] E. Zhou, Q. Su, C. Chi, Z. Zhang *et al.*, "Code-as-monitor: Constraint-aware visual programming for reactive and proactive robotic failure detection," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 6919–6929. [Online]. Available: https://doi.org/10.1109/CVPR52734.2025.00649

[9] R. Sinha, A. Elhafsi, C. Agia, M. Foutter, E. Schmerling, and M. Pavone, "Real-time anomaly detection and reactive planning with large language models," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[10] J. Yuan, Y. Wu, and A. Bajcsy, "When to act, ask, or learn: Uncertainty-aware policy steering," in *Proc. Robotics: Science and Systems (RSS)*, 2026, to appear (per arXiv v2 comments); proceedings entry not independently checked.

[11] A. Oloo, Z. Altaweel, Y. Hayamizu, P. Liu, Y. Ding, S. Amiri, H. Yang, A. Kaminski, C. Esselink, C. Paxton, X. Zhang, and S. Zhang, "Robot planning and situation handling with active perception," 2026, arXiv preprint; method name: VAP-TAMP.

[12] A. Elhafsi, R. Sinha, C. Agia, E. Schmerling, I. Nesnas, and M. Pavone, "Semantic anomaly detection with large language models," 2023, arXiv preprint.

[13] S. Whitehead, S. Petryk, V. Shakib, J. Gonzalez, T. Darrell, A. Rohrbach, and M. Rohrbach, "Reliable visual question answering: Abstain rather than answer incorrectly," in *European Conference on Computer Vision (ECCV)*, 2022.

[14] A. Z. Ren, A. Dixit, A. Bodrova, S. Singh *et al.*, "Robots that ask for help: Uncertainty alignment for large language model planners," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 661–682. [Online]. Available: https://proceedings.mlr.press/v229/ren23a.html

[15] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," 2021. [Online]. Available: https://arxiv.org/abs/2107.07511

[16] K. Liang, Z. Zhang, and J. F. Fisac, "Introspective planning: Aligning robots' uncertainty with inherent task ambiguity," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024. [Online]. Available: https://papers.nips.cc/paper_files/paper/2024/hash/8451a20c5a7e0ee5671dda28f7daf7f3-Abstract-Conference.html

[17] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *International Conference on Learning Representations (ICLR)*, 2024.

[18] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *The Annals of Applied Statistics*, vol. 19, no. 2, pp. 1641–1662, 2025.

[19] J. Lekeufack, A. N. Angelopoulos, A. Bajcsy, M. I. Jordan, and J. Malik, "Conformal decision theory: Safe autonomous decisions from imperfect predictions," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 11 668–11 675.

[20] B. Ichter, A. Brohan, Y. Chebotar, C. Finn *et al.*, "Do as I can, not as I say: Grounding language in robotic affordances," in *Proceedings of The 6th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2022, pp. 287–318, coRL 2022 (PMLR volume dated 2023). Author order follows the PMLR record; the arXiv version (2204.01691) lists Michael Ahn first ("Ahn et al., 2022"). [Online]. Available: https://proceedings.mlr.press/v205/ichter23a.html

[21] J. Robinson, C. M. Rytting, and D. Wingate, "Leveraging large language models for multiple choice question answering," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=yKbprarjc5B

[22] S. Kadavath, T. Conerly, A. Askell, T. Henighan *et al.*, "Language models (mostly) know what they know," 2022. [Online]. Available: https://arxiv.org/abs/2207.05221

[23] C. Zheng, H. Zhou, F. Meng, J. Zhou, and M. Huang, "Large language models are not robust multiple choice selectors," in *International Conference on Learning Representations (ICLR)*, 2024. [Online]. Available: https://openreview.net/forum?id=shr9PXz7T0

[24] P. Pezeshkpour and E. Hruschka, "Large language models sensitivity to the order of options in multiple-choice questions," in *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, 2024, pp. 2006–2017. [Online]. Available: https://doi.org/10.18653/v1/2024.findings-naacl.130

[25] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 70. PMLR, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[26] Z. Zhao, E. Wallace, S. Feng, D. Klein, and S. Singh, "Calibrate before use: Improving few-shot performance of language models," in *Proceedings of the 38th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 12 697–12 706. [Online]. Available: https://proceedings.mlr.press/v139/zhao21c.html

[27] T. Wu, C. Zhou, G. Zhao, H. Cao, Y. Pu, and J. Yang, "When robots should say "I don't know": Benchmarking abstention in embodied question answering," 2025, arXiv preprint.

[28] X. Wang, J. Wei, D. Schuurmans, Q. Le *et al.*, "Self-consistency improves chain of thought reasoning in language models," in *International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=1PL1NIMMrw

[29] Y. Li, Y. Du, K. Zhou, J. Wang *et al.*, "Evaluating object hallucination in large vision-language models," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2023, pp. 292–305. [Online]. Available: https://doi.org/10.18653/v1/2023.emnlp-main.20

[30] D. Seto, B. Krogh, L. Sha, and A. Chutinan, "The Simplex architecture for safe online control system upgrades," in *Proceedings of the 1998 American Control Conference (ACC)*. IEEE, 1998, pp. 3504–3508. [Online]. Available: https://doi.org/10.1109/ACC.1998.703255

[31] L. Sha, "Using simplicity to control complexity," *IEEE Software*, vol. 18, no. 4, pp. 20–28, 2001. [Online]. Available: https://doi.org/10.1109/MS.2001.936213

[32] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer *et al.*, "Safe reinforcement learning via shielding," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018. [Online]. Available: https://doi.org/10.1609/aaai.v32i1.11797

[33] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista *et al.*, "Control barrier functions: Theory and applications," in *18th European Control Conference (ECC)*, 2019, pp. 3420–3431. [Online]. Available: https://doi.org/10.23919/ECC.2019.8796030

[34] A. Robey, Z. Ravichandran, V. Kumar, H. Hassani, and G. J. Pappas, "Jailbreaking LLM-controlled robots," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 11 948–11 956. [Online]. Available: https://doi.org/10.1109/ICRA55743.2025.11128119

[35] Z. Ravichandran, A. Robey, V. Kumar, G. J. Pappas, and H. Hassani, "Safety guardrails for LLM-enabled robots," *IEEE Robotics and Automation Letters*, vol. 11, no. 4, pp. 4649–4656, 2026. [Online]. Available: https://doi.org/10.1109/LRA.2026.3667488

[36] K.-C. Hsu, H. Hu, and J. F. Fisac, "The safety filter: A unified view of safety-critical control in autonomous systems," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 7, no. 1, pp. 47–72, 2024.

[37] L. Santos, Z. Li, L. Peters, S. Bansal, and A. Bajcsy, "Updating robot safety representations online from natural language feedback," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 7778–7785.

[38] L. Wang, Z. Ying, X. Yang, Q. Zou, Z. Yin, T. Li, J. Yang, Y. Yang, A. Liu, and X. Liu, "RoboSafe: Safeguarding embodied agents via executable safety logic," 2025, arXiv preprint.

[39] B. Zitkovich, T. Yu, S. Xu, P. Xu *et al.*, "RT-2: Vision-language-action models transfer web knowledge to robotic control," in *Proceedings of The 7th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 2165–2183, author order follows the PMLR record; the arXiv version (2307.15818) lists Anthony Brohan first ("Brohan et al., 2023"). [Online]. Available: https://proceedings.mlr.press/v229/zitkovich23a.html

[40] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao *et al.*, "OpenVLA: An open-source vision-language-action model," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 2679–2713, coRL 2024 (PMLR volume dated 2025). [Online]. Available: https://proceedings.mlr.press/v270/kim25c.html

[41] A. Anwar, J. Welsh, J. Biswas, S. Pouya, and Y. Chang, "ReMEmbR: Building and reasoning over long-horizon spatio-temporal memory for robot navigation," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 2838–2845.

[42] Q. Xie, S. Y. Min, P. Ji, Y. Yang, T. Zhang, K. Xu, A. Bajaj, R. Salakhutdinov, M. Johnson-Roberson, and Y. Bisk, "Embodied-RAG:

General non-parametric embodied memory for retrieval and generation," 2024, arXiv preprint.

[43] Z. Wang, B. Yu, J. Zhao, W. Sun, S. Hou, S. Liang, X. Hu, Y. Han, and Y. Gan, "KARMA: Augmenting embodied AI agents with long-and-short term memory systems," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

[44] L. Bärmann, C. DeChant, J. Plewnia, F. Peller-Konrad, D. Bauer, T. Asfour, and A. Waibel, "Episodic memory verbalization using hierarchical representations of life-long robot experience," in *Proc. IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2025, pp. 783–790.

[45] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial perception system for 3D scene graph construction and optimization," in *Proc. Robotics: Science and Systems (RSS)*, 2022.

[46] Q. Gu, A. Kuwajerwala, S. Morin, K. M. Jatavallabhula, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. M. de Melo, J. B. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, "ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 5021–5028.

[47] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, "Hierarchical open-vocabulary 3D scene graphs for language-grounded robot navigation," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[48] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Suenderhauf, "SayPlan: Grounding large language models using 3D scene graphs for scalable robot task planning," in *Proc. Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229. PMLR, 2023, pp. 23–72.

[49] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 10 608–10 615. [Online]. Available: https://doi.org/10.1109/ICRA48891.2023.10160969

[50] P. Biber and T. Duckett, "Dynamic maps for long-term operation of mobile service robots," in *Proc. Robotics: Science and Systems (RSS) I*, 2005, pp. 17–24.

[51] T. Krajník, J. P. Fentanes, J. P. S. S. Santos, and T. Duckett, "FreMEn: Frequency map enhancement for long-term mobile robot autonomy in changing environments," *IEEE Transactions on Robotics*, vol. 33, no. 4, pp. 964–977, 2017.

[52] L. Schmid, M. Abate, Y. Chang, and L. Carlone, "Khronos: A unified approach for spatio-temporal metric-semantic SLAM in dynamic environments," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[53] T. Behrens, R. Zurbrügg, M. Pollefeys, Z. Bauer, and H. Blum, "Lost & found: Tracking changes from egocentric observations in 3D dynamic scene graphs," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 3739–3746, 2025.

[54] Y. Sun and Y. Zhang, "When memory lies: An empirical study of spatial memory staleness in VLM agents," 2026, arXiv preprint.

[55] Z. Chen, Z. Xiang, C. Xiao, D. Song, and B. Li, "AgentPoison: Red-teaming LLM agents via poisoning memory or knowledge bases," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

[56] K. Yadav, Y. Ali, G. Gupta, Y. Gal, and Z. Kira, "FindingDory: A benchmark to evaluate memory in embodied agents," 2025, arXiv preprint.

[57] X. Yin and H. Du, "GLOVE: Global verifier for LLM memory-environment realignment," 2026, arXiv preprint.

[58] G. Tang, Q. Jia, G. Chen, T. Li, Z. Huang, Z. Lv, and N. Ji, "VLM-DEWM: Dynamic external world model for verifiable and resilient vision-language planning in manufacturing," 2026, arXiv preprint.

[59] A. Aydemir, A. Pronobis, M. Göbelbecker, and P. Jensfelt, "Active visual object search in unknown environments using uncertain semantics," *IEEE Transactions on Robotics*, vol. 29, no. 4, pp. 986–1002, 2013.

[60] Z. Zeng, A. Röfer, and O. C. Jenkins, "Semantic linking maps for active visual object search," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2020, pp. 1984–1990, arXiv:2006.10807.

[61] A. Wandzel, Y. Oh, M. Fishman, N. Kumar, L. L. S. Wong, and S. Tellex, "Multi-object search using object-oriented POMDPs," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2019, pp. 7194–7200.

[62] J. Mason and B. Marthi, "An object-based semantic world model for long-term change detection and semantic querying," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 3851–3858.

[63] D. M. Rosen, J. Mason, and J. J. Leonard, "Towards lifelong feature-based mapping in semi-static environments," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2016, pp. 1063–1070.

[64] Z. Yan, S. Li, Z. Wang, L. Wu, H. Wang, J. Zhu, L. Chen, and J. Liu, "Dynamic open-vocabulary 3D scene graphs for long-term language-guided mobile manipulation," *IEEE Robotics and Automation Letters*, vol. 10, no. 5, pp. 4252–4259, 2025, arXiv:2410.11989.

[65] K. Zhou, K. Zheng, C. Pryor, Y. Shen, H. Jin, L. Getoor, and X. E. Wang, "ESC: Exploration with soft commonsense constraints for zero-shot object navigation," in *Proc. 40th Int. Conf. Machine Learning (ICML), PMLR 202*, 2023, pp. 42 829–42 842, arXiv:2301.13166.

[66] N. Yokoyama, S. Ha, D. Batra, J. Wang, and B. Bucher, "VLFM: Vision-language frontier maps for zero-shot semantic navigation," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2024, pp. 42–48, arXiv:2312.03275.

[67] R. Bajcsy, "Active perception," *Proceedings of the IEEE*, vol. 76, no. 8, pp. 966–1005, 1988.

[68] R. Bajcsy, Y. Aloimonos, and J. K. Tsotsos, "Revisiting active perception," *Autonomous Robots*, vol. 42, no. 2, pp. 177–196, 2018.

[69] W. R. Scott, G. Roth, and J.-F. Rivest, "View planning for automated three-dimensional object reconstruction and inspection," *ACM Computing Surveys*, vol. 35, no. 1, pp. 64–96, 2003.

[70] A. Z. Ren, J. Clark, A. Dixit, M. Itkina, A. Majumdar, and D. Sadigh, "Explore until confident: Efficient exploration for embodied question answering," in *Proc. Robotics: Science and Systems (RSS)*, 2024.

[71] V. Sripada, S. Carter, F. Guerin, and A. Ghalamzan, "Scene exploration by vision-language models," 2024, arXiv preprint.

[72] D. Hendrycks and T. Dietterich, "Benchmarking neural network robustness to common corruptions and perturbations," in *Proc. International Conference on Learning Representations (ICLR)*, 2019.

[73] C. Michaelis, B. Mitzkus, R. Geirhos, E. Rusak, O. Bringmann, A. S. Ecker, M. Bethge, and W. Brendel, "Benchmarking robustness in object detection: Autonomous driving when winter is coming," 2019, arXiv preprint.

[74] S. Chen, J. Gu, Z. Han, Y. Ma, P. Torr, and V. Tresp, "Benchmarking robustness of adaptation methods on pre-trained vision-language models," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023.

[75] M. C. Schiappa, S. Vyas, H. Palangi, Y. S. Rawat, and V. Vineet, "Robustness analysis of video-language models against visual and language perturbations," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 35, 2022.

[76] S. Höfer, K. Bekris, A. Handa, J. C. Gamboa, M. Mozifian, F. Golemo *et al.*, "Sim2Real in robotics and automation: Applications and challenges," *IEEE Transactions on Automation Science and Engineering*, vol. 18, no. 2, pp. 398–400, 2021.

[77] A. Avižienis, "The N-version approach to fault-tolerant software," *IEEE Transactions on Software Engineering*, vol. SE-11, no. 12, pp. 1491–1501, 1985.

[78] J. C. Knight and N. G. Leveson, "An experimental evaluation of the assumption of independence in multiversion programming," *IEEE Transactions on Software Engineering*, vol. SE-12, no. 1, pp. 96–109, 1986.

[79] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning in language models through multiagent debate," in *Proc. International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 11 733–11 763.

[80] H. Kress-Gazit, K. Hashimoto, N. Kuppuswamy, P. Shah *et al.*, "Robot learning as an empirical science: Best practices for policy evaluation," 2024. [Online]. Available: https://arxiv.org/abs/2409.09491

[81] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927. [Online]. Available: https://doi.org/10.1080/01621459.1927.10502953

[82] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947. [Online]. Available: https://doi.org/10.1007/BF02295996

[83] J. Tobin, R. Fong, A. Ray, J. Schneider *et al.*, "Domain randomization for transferring deep neural networks from simulation to the real world," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30. [Online]. Available: https://doi.org/10.1109/IROS.2017.8202133

[84] D. Snyder, A. J. Hancock, A. Badithela, E. Dixon, P. Miller, R. A. Ambrus, A. Majumdar, M. Itkina, and H. Nishimura, "Is your imitation

learning policy better than mine? policy comparison with near-optimal stopping," in *Robotics: Science and Systems XXI (RSS)*, 2025.

[85] X. Li, K. Hsu, J. Gu, K. Pertsch, O. Mees *et al.*, "Evaluating real-world robot manipulation policies in simulation," in *Proceedings of The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2024, pp. 3705–3728, coRL 2024 (PMLR volume dated 2025); method name: SIMPLER.

[86] P. V. A. B. Venâncio, A. C. Lisboa, and A. V. Barbosa, "An automatic fire detection system based on deep convolutional neural networks for low-power, resource-constrained devices," *Neural Computing and Applications*, vol. 34, no. 18, pp. 15 349–15 368, 2022.

[87] M. Wang, P. Yue, L. Jiang, D. Yu, T. Tuo, and J. Li, "An open flame and smoke detection dataset for deep learning in remote sensing based fire detection," *Geo-spatial Information Science*, vol. 28, no. 2, pp. 511–526, 2025.

[88] L. Seidel, S. Gehringer, T. Raczok, S.-N. Ivens, B. Eckardt, and M. Maerz, "Advancing early wildfire detection: Integration of vision language models with unmanned aerial vehicle remote sensing for enhanced situational awareness," *Drones*, vol. 9, no. 5, p. 347, 2025.

[89] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 11 975–11 986.

[90] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 139, 2021, pp. 8748–8763.

TABLE S8

CAMPAIGN H, ALL PAIRED MISSIONS (SEQ = SEQUENCE.MISSION). OUTCOME AND MISSION TIME: ✓ DELIVERED, NO UNSAFE DELIVERY; × UNSAFE DELIVERY; ○ REFUSED WHERE NO SAFE OPTION EXISTED; △ REFUSED ALTHOUGH A SAFE OPTION EXISTED (FALSE REFUSAL); NO MISSION FAILED. TRANSITION: WHAT CHANGED SINCE THE PREVIOUS MISSION. STORIES (MISSIONS 1→2→3): S1 NOMINAL, SWAPPED, SWAPPED; S2 SWAPPED, SWAPPED, NOMINAL; S3 NO FIRE, FIRE BEHIND THE REQUESTED STATION, FIRE; S4 FIRE, NO FIRE, FIRE. M2 SOURCE: FIRST STATION TAKEN FROM MEMORY AND/OR CAUTION APPLIED TO A STATION.

Seq	Story	Transition	M0	M1	M2	M2 source
Q01.2	S3	fire appears	○ (62 s)	× (41 s)	○ (45 s)	memory
Q01.3	S3	fire persists	○ (30 s)	× (41 s)	○ (30 s)	memory+caution
Q02.2	S2	relocation persist	✓(112 s)	✓(28 s)	✓(72 s)	memory
Q02.3	S2	relocation change	✓(85 s)	× (28 s)	✓(114 s)	memory
Q03.2	S1	relocation change	✓(111 s)	× (28 s)	✓(114 s)	memory
Q03.3	S1	relocation persist	✓(115 s)	× (28 s)	✓(86 s)	memory
Q04.2	S4	fire gone	✓(84 s)	△ (3 s)	✓(83 s)	memory+caution
Q04.3	S4	fire appears	○ (44 s)	○ (3 s)	○ (44 s)	memory+caution
Q05.2	S3	fire appears	○ (59 s)	× (29 s)	○ (59 s)	memory
Q05.3	S3	fire persists	○ (43 s)	× (28 s)	○ (41 s)	memory+caution
Q06.2	S4	fire gone	✓(83 s)	△ (3 s)	✓(83 s)	memory+caution
Q06.3	S4	fire appears	○ (31 s)	○ (3 s)	○ (32 s)	memory+caution
Q07.2	S1	relocation change	✓(108 s)	× (42 s)	✓(109 s)	memory
Q07.3	S1	relocation persist	✓(115 s)	× (39 s)	✓(72 s)	memory
Q08.2	S4	fire gone	✓(74 s)	△ (3 s)	✓(72 s)	memory+caution
Q08.3	S4	fire appears	○ (28 s)	○ (3 s)	○ (29 s)	memory+caution
Q09.2	S4	fire gone	✓(83 s)	△ (3 s)	✓(85 s)	memory+caution
Q09.3	S4	fire appears	× (85 s)	○ (3 s)	○ (44 s)	memory+caution
Q10.2	S2	relocation persist	✓(106 s)	✓(27 s)	✓(72 s)	memory
Q10.3	S2	relocation change	✓(83 s)	× (27 s)	✓(113 s)	memory
Q11.2	S2	relocation persist	✓(114 s)	✓(39 s)	✓(85 s)	memory
Q11.3	S2	relocation change	✓(74 s)	× (39 s)	✓(106 s)	memory
Q12.2	S3	fire appears	○ (42 s)	× (29 s)	○ (41 s)	memory
Q12.3	S3	fire persists	○ (29 s)	× (29 s)	○ (28 s)	memory+caution
Q13.2	S3	fire appears	○ (29 s)	× (29 s)	○ (28 s)	memory
Q13.3	S3	fire persists	○ (62 s)	× (28 s)	○ (28 s)	memory+caution
Q14.2	S2	relocation persist	✓(120 s)	✓(40 s)	✓(85 s)	memory
Q14.3	S2	relocation change	✓(75 s)	× (40 s)	✓(113 s)	memory
Q15.2	S1	relocation change	✓(125 s)	× (28 s)	✓(116 s)	memory
Q15.3	S1	relocation persist	✓(114 s)	× (28 s)	✓(96 s)	memory
Q16.2	S1	relocation change	✓(109 s)	× (41 s)	✓(112 s)	memory
Q16.3	S1	relocation persist	✓(108 s)	× (41 s)	✓(73 s)	memory

TABLE S7

WHEN EACH STUDY WAS REGISTERED AND WHEN ITS FIRST RESULTS ENTERED THE REPOSITORY. "ORDER" CHECKS THAT THE REGISTRATION COMMIT IS AN ANCESTOR OF THE FIRST RESULTS COMMIT (`GIT MERGE-BASE --IS-ANCESTOR`). REGISTRATIONS ARE ALSO MIRRORED IN THE PUBLIC REPOSITORY HISTORY.

Study	Registered (commit, date)	First results (commit, date)	Amendments	Order
C (v3.3)	`5c9a951 2026-10-16`	`5c9a951 2026-10-16`	0	same commit
D (v3.4)	`ac6d815 2026-10-16`	`fb4e9b3 2026-10-16`	0	✓
T1	`5f835af 2026-10-16`	`c3d1ffc 2026-10-16`	3	✓
E	`6201aa0 2026-10-16`	`17e0ea6 2026-10-16`	0	✓
T2	`f347239 2026-10-17`	`7843e9c 2026-10-17`	1	✓
F	`564a724 2026-10-17`	`5bbf11a 2026-10-17`	0	✓
G	`edb26a1 2026-10-17`	`f6160fb 2026-10-17`	1	✓
R	`fb63bee 2026-10-17`	`2337e49 2026-10-17`	0	✓
H	`a5cc651 2026-10-17`	`389ffc8 2026-10-17`	1	✓
J	`b72b877 2026-10-17`	`pending`	0	pending
V	`b23210f 2026-10-17`	`pending`	1	pending
W	`b2a6dbe 2026-10-17`	`pending`	0	pending
Z	`a53ee7e 2026-10-17`	`0ca401d 2026-10-17`	0	✓
K	`3a21238 2026-10-17`	`9b2086f 2026-10-17`	0	✓
P	`e6bde02 2026-10-17`	`00795a8 2026-10-17`	0	✓
P2	`72f0309 2026-10-17`	`pending`	1	pending